

Exploring the Role of Economic Integration in Inequality: A Fixed Effects and First Difference Analysis of Trade Openness and the Gini Index

ECON 114: Advanced Quantitative Methods

Louisa Gallagher

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This is a condensed version of the original paper. Figures and tables are included at the end of the paper.

Abstract

This paper employs fixed effects and first difference models to examine the relationship between economic integration, represented by trade openness, and inequality. Panel data from the World Bank DataBank is used to look at the short- and longer-term relationship between 1996 and 2019. Using the small set of countries with complete data, fixed effects and first difference specifications are estimated, taking into account time-invariant country characteristics. Due to concerns of heteroscedasticity and serial correlation, inference is conducted through bootstrapping and clustering at the country level. While the resulting estimates are positive, they are practically and statistically insignificant. Thus, the results provide limited evidence of a non-zero relationship between trade openness and the Gini index. Potential bias may be a concern as a result of the small number of units and time-varying country characteristics that are not accounted for in the models.

I Introduction

The recent threats and implementations of tariffs have ignited larger conversations around the potential effects of these measures, typically regarding the impact on employment, prices, investment, and industries. As recently observed, these tariffs are often accompanied by retaliatory measures, affecting multiple trading partners and potentially reducing economic integration. This raises broader questions about the impact of economic integration on economic outcomes within countries. This paper will therefore examine the relationship between trade openness and income inequality, measured using the Gini index.

Using panel data from the World Bank's World Development Indicators for 1996–2019, this paper employs fixed effects and first difference models to account for time-invariant country characteristics that may affect both trade openness and inequality. The analysis examines both short- and longer-term relationships by considering one- and five-year lags of trade openness and the control variables. Models are estimated with and without control variables for robustness. Due to concerns of heteroscedasticity and serial correlation, inference accounts for these concerns through bootstrapping and clustering at the country level. A key limitation is the small number of countries with complete data, which may introduce selection bias.

The resulting estimates are generally positive, yet small in magnitude, and statistical significance varies across specifications and inference methods. The results therefore do not provide robust evidence of a relationship between trade openness and inequality. The limited explanatory power of trade openness may indicate that more specific factors influence inequality, and that an aggregate measure of inequality may not properly capture these effects.

II Data

The data is drawn from the World Bank's World Development Indicators database (World Bank, 2025), providing information for 217 economies. This analysis focuses on the period from 1996 to 2019, beginning just after the creation of the World Trade Organization in 1995, and ending before the COVID-19 pandemic which disrupted global trade.

II.1 Variables

To examine the relationship between economic integration and inequality, economic integration will be represented by trade openness, measured as the sum of a country's imports and exports of goods and services as a percentage of their GDP. The Gini index will be used as

the measure of inequality, which ranges from 0, perfect equality, to 100, perfect inequality. Factors other than trade that may affect inequality are included as control variables. Unemployment is measured using the modeled ILO estimate for consistency across countries, while GDP per capita is measured in constant 2015 U.S. dollars to ensure comparability across countries and over time. Inflation, measured using annual consumer prices, is included because changes in purchasing power may affect income groups differently. Finally, tax revenue, measured as a percentage of GDP, is included as an indicator of fiscal space which may affect a country's ability to address inequality.

II.2 Complete Data

In order to perform the analysis, countries without complete data for any variables were excluded, resulting in a small subset of 8 countries. This small sample size is an important consideration when performing inference and interpreting results. For the final sample of countries, Table 1 presents the descriptive statistics of each variable in the data set, while Figure 1 shows the visual trends in these variables over time for each country.

Another concern is why countries may lack data for certain variables. In many cases, political factors or limitations in infrastructure and resources may hinder a country's ability to measure and report certain indicators. These factors may also affect trade openness and inequality. As a result, selection bias may be present in the results, as the sample of data is representative only of countries with the ability to measure and report the necessary data.

II.3 Temporal Dynamics

Due to the possibility that changes in trade and the control variables may take time to affect inequality, trade and the controls are lagged. Inequality may adjust gradually as countries and industries respond to changing economic conditions, and the short- and longer-term relationships may differ. To retain as much data as possible, this paper examines a lag of 1 year to represent a short-term relationship, and a lag of 5 years to capture a longer-term relationship.

III Empirical Methods

The panel data structure allows for the examination of changes in trade openness and inequality over time. Fixed effects and first difference models will therefore be estimated and compared, with both focusing on within-country variation.

III.1 Fixed Effects Model

The fixed effects model takes into account unobserved country specific characteristics that are constant over time, such as geography, climate, trade agreements, and historical connections which may affect trade openness and inequality. It also accounts for time fixed effects that are constant across countries, which capture global trends that affect trade and inequality globally, such as the Global Financial crisis in 2008. The fixed effects specifications will therefore take the form of Equation (1).

$$Gini_{it} = \alpha_0 + \beta_1 trade_{it} + \beta'X_{it} + \alpha_i + \gamma_t + u_{it} \quad (1)$$

The estimate of interest is β_1 which captures the within-country relationship between trade openness and the Gini index after accounting for the country fixed effects, α_i , and the time fixed effects, γ_t . X_{it} represents a vector of time-varying control variables. Comparing specifications with and without these controls will provide a check on the robustness of the estimate. The error term, u_{it} , captures the variation in inequality that is unexplained by trade, the controls, and the fixed effects. The main assumption of the fixed effects model is that this error term is uncorrelated with trade openness within countries.

III.2 First Difference Model

The first difference model will similarly help to eliminate time-constant and unobserved country specific characteristics by differencing the variables to focus on within-country changes over time. The first difference specifications will therefore take the form of Equation (2).

$$\Delta Gini_{it} = \beta_1 \Delta trade_{it} + \beta' \Delta X_{it} + \Delta u_{it} \quad (2)$$

The estimate of interest will be the effect of changes in trade openness on changes in the Gini index, captured by β_1 . ΔX_{it} represents a vector of the changes in the same time-varying control variables used in the fixed effects specification. Finally, Δu_{it} is the change in the error term that captures the additional unexplained variation in the changes of inequality. The main assumption is that changes in trade are uncorrelated with changes in the error term.

III.3 Inference

Heteroscedasticity and serial correlation are potential concerns due to variation in the error term across countries and time, and the likelihood that observations and errors may be correlated over time within the same country. This arises due to looking at factors like trade openness and inequality that are persistent and only gradually change due to preexisting or

previous conditions. For these reasons, the naive standard errors should not be relied upon. Instead, bootstrapping and clustering will be used to construct 95% confidence intervals. The bootstrapping method entails 10,000 resamples of the data, with replacement, producing 10,000 bootstrap estimates. Confidence intervals are then constructed based on the 2.5% and 97.5% quartiles of the bootstrap distribution. These will be compared to the confidence intervals constructed using the clustered standard errors. Clustering will be performed at the country level, allowing for correlation across time within the same country, although inference may be less reliable given the small number of countries with complete data.

IV Results

IV.1 Fixed Effects Estimates

The short-term fixed effects estimates of the effect of trade openness are presented in Table 2. Both specifications with and without controls suggest that a 10 percentage point increase in trade openness is associated with an increase of 0.19 points in the Gini index. The naive standard errors indicate that these estimates are statistically significant at the 1% and 5% levels, although this does not account for serial correlation and heteroskedasticity. This estimated effect is also very small, especially relative to the scale of the Gini index.

For the longer-term relationship, the fixed effects estimates are presented in Table 3. Here the impact of a 10 percentage point increase in trade openness on the Gini index is estimated to be 0.37 without controls, and 0.43 with controls. Both estimates are significant at the 1% significance level according to the naive standard errors. The estimate increases slightly in magnitude with the addition of controls, which could reflect omitted variable bias if the included controls affect both trade openness and inequality which influences the estimated coefficient when they are excluded. For example, if tax revenue is negatively associated with inequality while being positively associated with trade openness, omitting it could push the trade coefficient downward. However, other controls may bias the estimate in the opposite direction. While these coefficients are larger than the short-term estimates, they still remain very small in magnitude.

IV.2 First Difference Estimates

The first difference estimates without controls are presented in the first columns of Tables 4 and 5. The one-year lag specification without controls suggests that a 10 percentage point increase in trade sees an increase in the Gini index of 0.23. This is significant at the 5% level using naive standard errors, and the estimate is similar in magnitude to the corresponding fixed

effects estimate of 0.19 from Table 2. With a lag of 5 years, the model estimates a smaller change of 0.12 in the Gini index, contrasting with the fixed effects estimate of 0.37 in Table 3.

Column 2 of Tables 4 and 5 present the estimates from the specifications with the same controls as the fixed effects model. These effects are both statistically insignificant according to the naive standard errors, and much smaller than the specifications without controls. This difference may reflect the fact that the controls are time-varying, while the first difference model focuses on changes within countries over time. Controlling for changes in these variables may therefore reduce the variation in trade openness used to estimate its relationship with inequality, resulting in smaller estimates. The five-year lag estimate with controls is also negative, however it is extremely small and very close to 0 on the scale of the Gini index.

IV.3 Inference - Bootstrap and Clustered Confidence Intervals

Figure 2 presents the bootstrap distribution for the fixed effects and first difference estimates with and without controls using a lag of 1 year. For all four estimates, the distributions appear to be roughly centered around their respective original estimates, although the fixed effects distributions appear to have slightly longer left tails. This could be caused by limited variation in the data due to the small sample, and thus countries with smaller values could be over-represented when resampling. The 95% bootstrap confidence intervals in Table 6 contain 0 for all four of these specifications, indicating that zero cannot be ruled out under bootstrap inference. Figure 3 and Table 7 present the corresponding results for the five-year lag. The distributions are again centered around the original estimates and are more consistent. The 95% confidence intervals again contain 0 for three of the four specifications. Overall, the bootstrap results do not provide evidence of a statistically significant relationship between trade openness and inequality in the short- or longer-term, as zero cannot be ruled out for most specifications.

Tables 8 and 9 present the clustered standard errors and the corresponding 95% confidence intervals are presented in Tables 10 and 11. For the one-year lag, all estimates, apart from the uncontrolled first difference estimate, have confidence intervals that contain 0. For a lag of 5 years, the estimates for all specifications, apart from the first difference model with controls, have confidence intervals that exclude 0. The clustered inference may therefore provide stronger evidence of a non-zero relationship in the longer-term.

A summary of the statistical significance of the estimates across inference methods is presented in Table 12. The naive standard errors result in more statistically significant estimates

in comparison to the bootstrap and clustered standard errors, consistent with the naive standard errors failing to account for heteroskedasticity and serial correlation. Statistical significance also varies between the bootstrap and clustered standard errors, indicating that statistical significance is sensitive to how uncertainty is estimated. For both methods, the limited sample size is an important consideration. Bootstrap resampling from only eight countries provides limited information about the sampling distribution, while clustered inference relies on only eight clusters, making the estimated standard errors potentially less reliable. Overall, the results provide limited and non-robust evidence of a relationship between trade openness and inequality, and zero cannot be ruled out across the full set of specifications.

IV.4 Discussion

The various specifications generally estimated a positive relationship between trade and the Gini index, suggesting that greater trade openness is associated with greater inequality. The estimated effect of a 10 percentage point increase in trade openness on the Gini index is on average 0.23 across all specifications. Given that the Gini index ranges from 0 to 100, the estimates are practically insignificant, especially in relation to a 10 percentage point increase in trade openness which would be a substantial change in trade flows relative to a country's GDP. These estimates may also be affected by selection bias, as the sample is restricted to a small number of countries with complete data, and the factors which affect data availability may also be related to trade openness and inequality.

Additionally, while the fixed effects and first difference models account for time-invariant country characteristics, time-varying factors may still bias the estimates. The fixed effects specification requires the error term to be uncorrelated with trade openness after accounting for the fixed effects and controls, while the first difference specification requires changes in the error term to be uncorrelated with changes in trade openness. These assumptions may be violated by time-varying country characteristics associated with both trade openness and inequality. Such country characteristics may include technological and productivity improvements, or changes in comparative advantage, which can affect both trade patterns and inequality through changes in industries and employment. The institutional capacities of countries and political environments may also evolve over time, affecting their ability to form trade agreements and maintain international relationships, as well as their ability or willingness to address inequality through redistribution of tax revenue to support their population. The

persistence of inequality may also limit the ability of the models to capture relationships over shorter periods, as the Gini index may adjust gradually. This may be particularly relevant for the first difference specifications, which rely on within-country variation.

Given the inability to reject that the relationship between trade openness and the Gini index is different from 0, trade openness appears to be a weak explanatory variable. Trade openness is simply the sum of a country's imports and exports as a percentage of GDP, and does not capture the composition of trade, nor how gains or losses from trade may be distributed among a population or industries. Inequality could be affected differently based on whether or not this trade is driven by imports or exports, if it is trade in goods or services, or the types of things traded. Thus, inequality may not be simply dependent on the level of trade openness, but more the specific characteristics of a country's trade and integration in the global economy.

V Conclusion

The varying inference results and practically insignificant estimates may suggest that trade openness has limited explanatory power for inequality across specifications and time periods. After comparing specifications and inference methods, the magnitudes of the resulting estimates were negligible on the scale of the Gini index, and statistical significance varied depending on the method used. Several factors may help explain these results. The small number of countries with complete data may introduce selection bias, meaning that the sample is not widely representative of all countries. Time-varying country characteristics that may be unobserved or difficult to control for are also a concern, as they may be correlated with trade openness and inequality, and thus violate the main assumptions of the models. Although the first difference and fixed effects models control for or eliminate time-invariant country characteristics and global time trends, time-varying country characteristics may still affect the estimates.

If trade openness alone is a weak explanatory variable, then this may motivate further research into the impact the composition of trade has on inequality, such as if increased imports or exports are found to have varying effects. Additional focus on specific trade policies may also serve to be more fruitful in estimating their specific impacts on society, as opposed to an aggregate measure like trade openness, which may lack a specificity needed to capture effects on societal conditions. Changes in inequality may instead reflect more complex economic and institutional dynamics, while the Gini index itself may be too broad to capture how these dynamics affect different groups within a population.

References

World Bank. (2024). World Development Indicators. World Bank Group. Retrieved March 6, 2025, from <https://databank.worldbank.org/source/world-development-indicators>

Table 1: Descriptive Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Trade (% GDP)	192	90.91	80.19	22.29	382.35
Gini Index	192	36.18	5.83	28.00	51.80
Unemployment Rate	192	7.78	3.58	1.80	20.71
GDP per Capita	192	40,090.01	28,028.83	1,187.12	112,417.90
Inflation Rate	192	3.20	4.16	-0.94	39.36
Tax Revenue (% GDP)	192	17.31	6.12	7.88	26.64

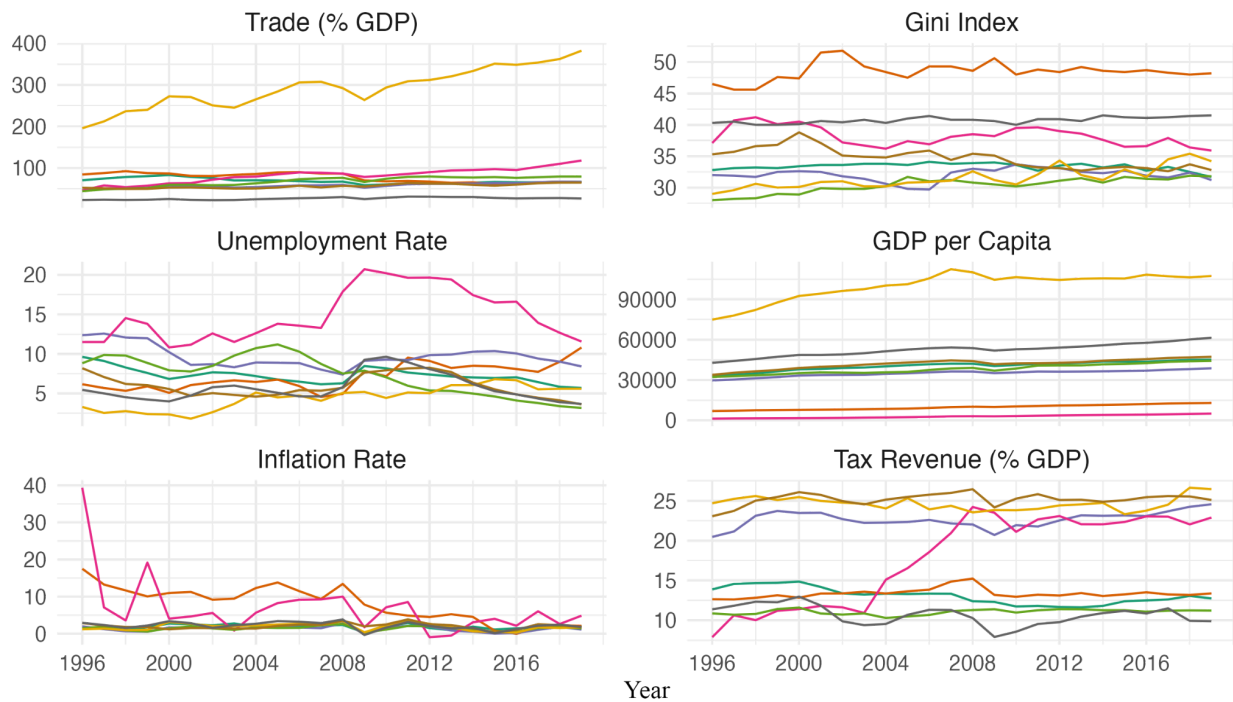
Figure 1: Variables Over Time

Table 2: Fixed Effects Regression Results (1 Year Lag)

	<i>Dependent variable:</i>	
	Gini (1)	Gini with Controls (2)
Trade (% GDP)	0.019*** (0.006)	0.019** (0.008)
Gini Index		0.058 (0.065)
Unemployment Rate		0.00001 (0.00005)
GDP per Capita		0.042 (0.034)
Inflation Rate		-0.148** (0.058)
Observations	184	184
R ²	0.064	0.144
Adjusted R ²	-0.119	-0.051
F Statistic	10.517*** (df = 1; 153)	5.028*** (df = 5; 149)

Note: Naïve standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 3: Fixed Effects Regression Results (5 Year Lag)

	<i>Dependent variable:</i>	
	Gini (1)	Gini with Controls (2)
Trade (% GDP)	0.037*** (0.006)	0.043*** (0.008)
Gini Index		-0.126** (0.056)
Unemployment Rate		-0.00002 (0.00004)
GDP per Capita		0.027 (0.027)
Inflation Rate		-0.032 (0.051)
Observations	152	152
R ²	0.241	0.312
Adjusted R ²	0.083	0.141
F Statistic	39.675*** (df = 1; 125)	10.957*** (df = 5; 121)

Note: Naïve standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 4: First Difference Regression Results (1 Year Lag)

	<i>Dependent variable:</i>	
	Gini (1)	Gini with Controls (2)
Trade (% GDP)	0.023** (0.010)	0.008 (0.012)
Gini Index		0.007 (0.074)
Unemployment Rate		0.0002** (0.0001)
GDP per Capita		0.014 (0.020)
Inflation Rate		0.148* (0.082)
Tax Revenue (% GDP)	-0.026 (0.072)	-0.103 (0.078)
Observations	176	176
R ²	0.028	0.078
Adjusted R ²	0.022	0.051
F Statistic	5.005** (df = 1; 174)	2.878** (df = 5; 170)

Note: Naive standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Table 5: First Difference Regression Results (5 Year Lag)

	<i>Dependent variable:</i>	
	Gini (1)	Gini with Controls (2)
Trade (% GDP)	0.012 (0.010)	-0.0001 (0.013)
Gini Index		-0.023 (0.078)
Unemployment Rate		0.0001* (0.0001)
GDP per Capita		0.018 (0.020)
Inflation Rate		-0.076 (0.086)
Tax Revenue (% GDP)	-0.077 (0.077)	-0.121 (0.086)
Observations	144	144
R ²	0.009	0.047
Adjusted R ²	0.002	0.013
F Statistic	1.289 (df = 1; 142)	1.376 (df = 5; 138)

Note: Naive standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1

Figure 2: Bootstrap Distribution of Estimates (1 Year Lag)

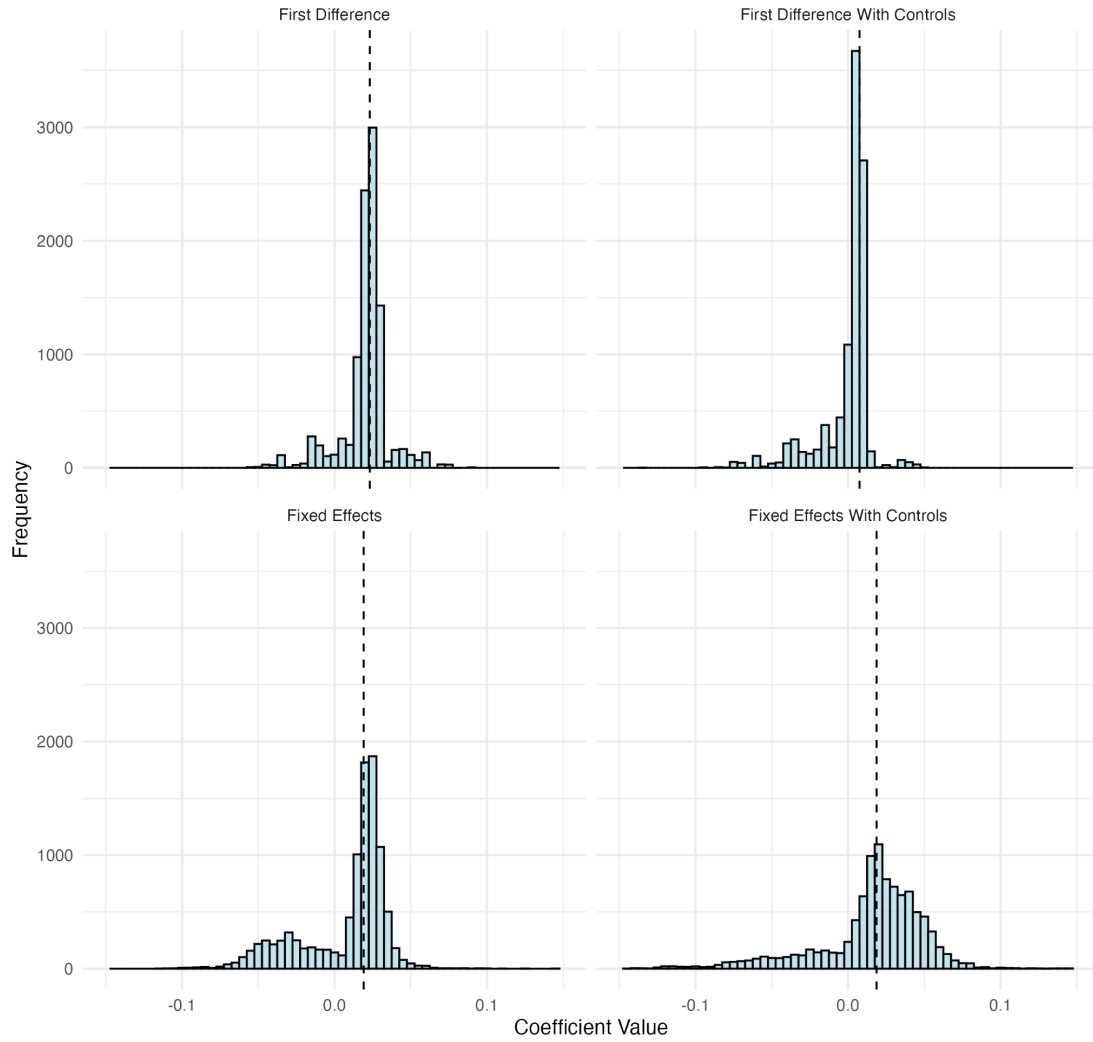


Table 6: Bootstrap Confidence Intervals (1 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	-0.058	0.041
Fixed Effects With Controls	-0.077	0.068
First Difference	-0.017	0.054
First Difference With Controls	-0.049	0.016

Table 7: Bootstrap Confidence Intervals (5 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	0.012	0.052
Fixed Effects With Controls	-0.0001	0.072
First Difference	-0.019	0.017
First Difference With Controls	-0.018	0.035

Figure 3: Bootstrap Distribution of Estimates (5 Year Lag)

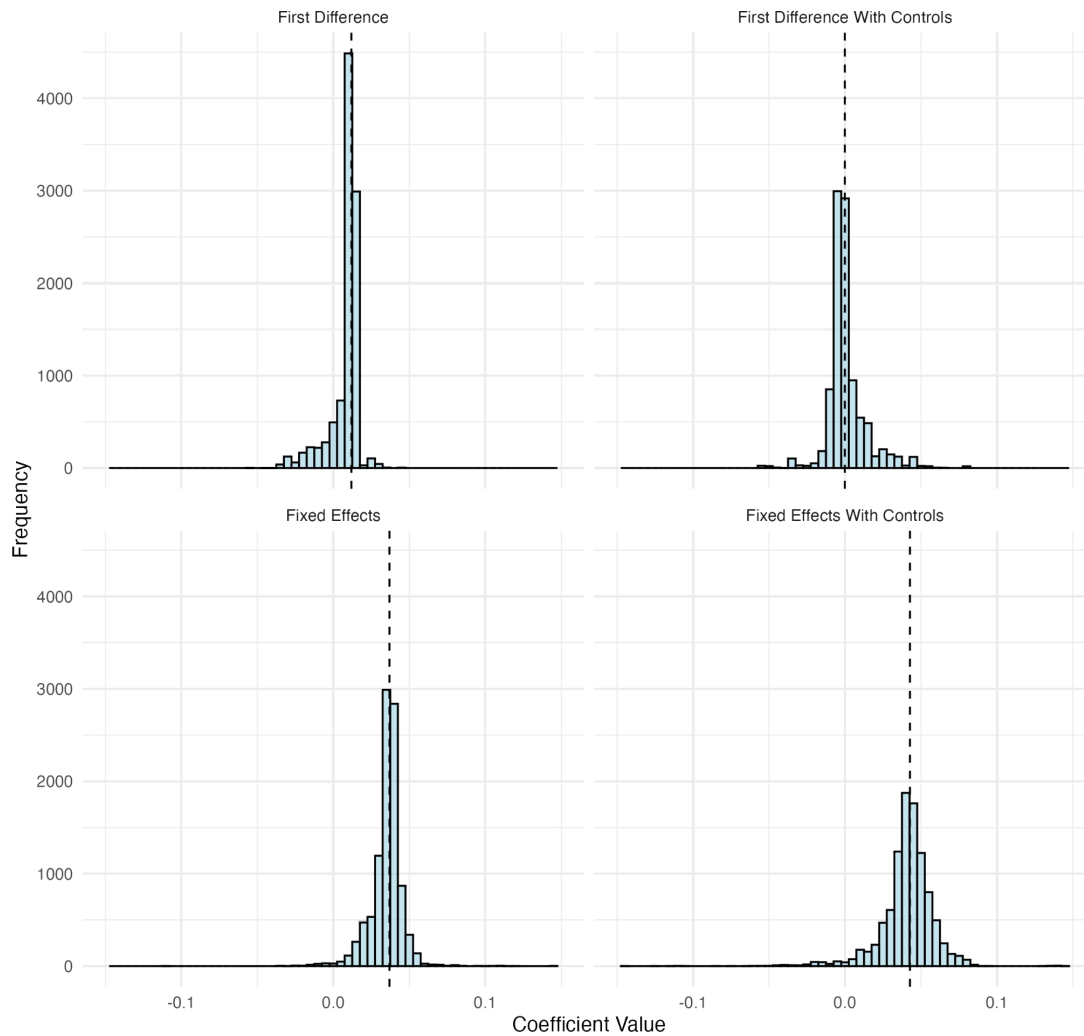


Table 8: Clustered Standard Errors (1 Year Lag)

Model	Standard Error
Fixed Effects	0.012
Fixed Effects With Controls	0.015
First Difference	0.007
First Difference With Controls	0.008

Table 9: Clustered Standard Errors (5 Year Lag)

Model	Standard Error
Fixed Effects	0.004
Fixed Effects With Controls	0.007
First Difference	0.004
First Difference With Controls	0.008

Table 10: Clustered Confidence Intervals (1 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	-0.004	0.042
Fixed Effects With Controls	-0.011	0.048
First Difference	0.010	0.036
First Difference With Controls	-0.009	0.024

Note: The confidence intervals are calculated using the standard errors presented in Table 8 and follow the standard equation of $\hat{\beta} \pm 1.96 \cdot SE(\hat{\beta})$.

Table 11: Clustered Confidence Intervals (5 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	0.029	0.045
Fixed Effects With Controls	0.030	0.055
First Difference	0.004	0.020
First Difference With Controls	-0.015	0.015

Note: The confidence intervals are calculated using the standard errors presented in Table 9 and follow the standard equation of $\hat{\beta} \pm 1.96 \cdot SE(\hat{\beta})$.

Table 12: Statistical Significance Summary

Model	Naïve Standard Errors	Bootstrap Standard Errors	Clustered Standard Errors
Fixed Effects (1 Year Lag)	Yes	No	No
Fixed Effects with Controls (1 Year Lag)	Yes	No	No
Fixed Effects (5 Year Lag)	Yes	Yes	Yes
Fixed Effects with Controls (5 Year Lag)	Yes	No	Yes
First Difference (1 Year Lag)	Yes	No	Yes
First Difference with Controls (1 Year Lag)	No	No	No
First Difference (5 Year Lag)	No	No	Yes
First Difference with Controls (5 Year Lag)	No	No	No

Note: “Yes” indicates that the coefficient on trade openness is statistically significant at the 5% level based on the 95% confidence intervals.